Extended Essay

Environmental Systems and Societies

The decrease in water level of the Dead Sea, Israel

Research question: To what extent do the inputs and outputs of water at the Dead Sea, Israel, allow a sustainable withdrawal of water by the chemical industry from 2000 to 2020?

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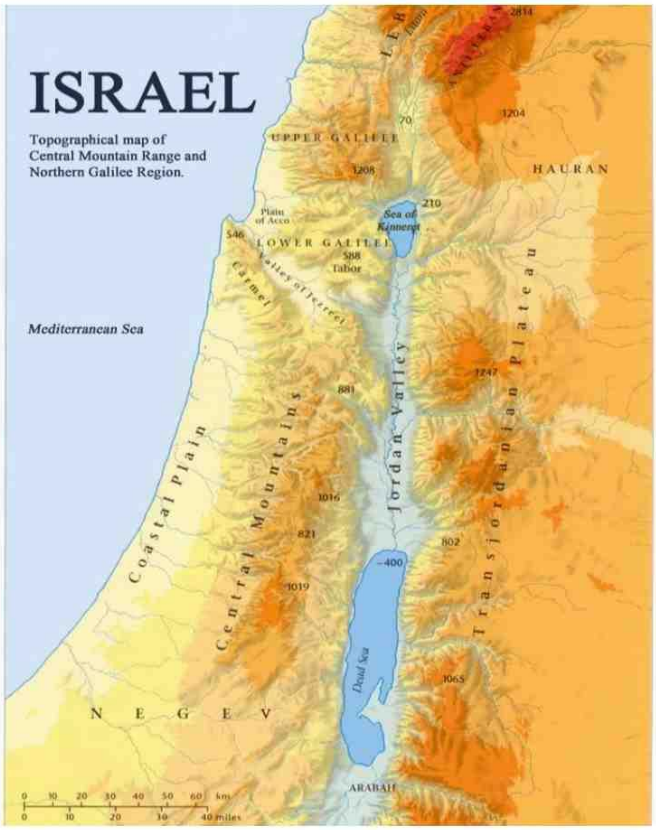
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Introduction

Background Information

The Dead Sea is the lowest place on earth and is located on the border between Israel and Jordan. It is characterised by its unique salty water composition, which is the main reason for the extensive tourism and the cultural importance of it. The high salinity of the water allows people to float on the water easily as the water density is higher, and the minerals and salts in it have health benefits which are another tourist attraction ("What Can We Do To Stop"). The salinity also allows no fish or other big organisms to live in the water, hence its name: Dead Sea. However, the water composition, that includes among other things high concentration of magnesium, allowed the development of a hot spot for microorganisms that live in craters at the bottom of the sea (Dell'Amore). These microorganisms are called halophiles (in Greek: salt-loving) and belong to the domain of Archaea (Ronel).

Figure 1: Topographic map of Israel and the Dead Sea



Source: Langfan, Mark. "Topographical Map of Israel." Israel Science and Technology Directory

The Dead Sea is located in the Judean Desert, Israel, which is naturally a dry place with low levels of precipitation and is classified as a rain shadow desert. Rain shadow desert is a natural phenomenon caused by the presence of mountains between a lower area and the sea, as can be seen from figure 1. When humid air comes from the Mediterranean Sea (at the west) it rains mostly in the Judaean (Central) Mountains. When it goes past the mountains, it reaches the Judaean desert, the temperature rises and the water evaporates, so that the volume of rain reaching the lower area of the desert is relatively small.

During the last few decades, the water level of the Dead Sea has been decreasing at an alarming rate of over 1 meter per year (Porat). The consequences of this process include the formation of sinkholes, depressions opened suddenly due to the dissolving of salt layers below the surface, and are surrounding the sea, and the destruction of the natural beaches.

Reasons for the Drying Process

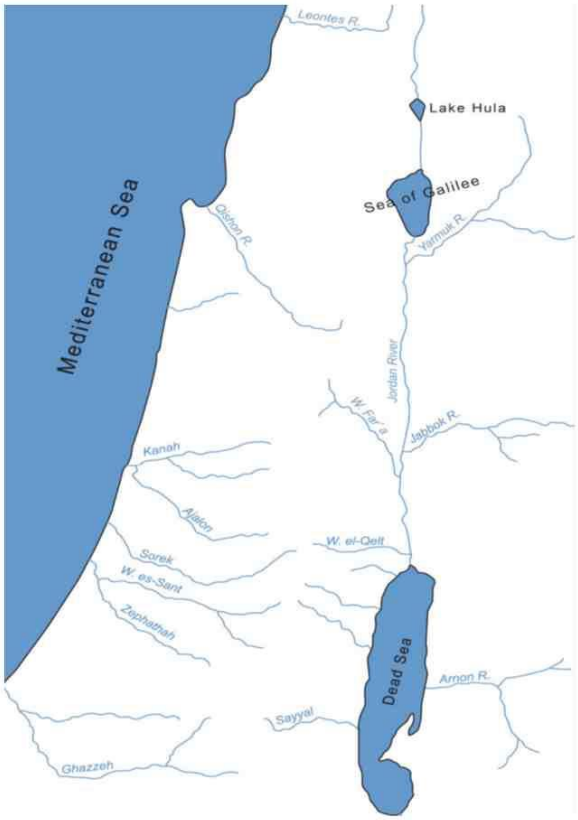
Water Outputs and Inputs Balance

There are several water inputs and outputs in the Dead Sea. The inputs include rainwater and surface water flow, which come from rivers, streams, springs, and floods that are created due to the impermeable desert soil which does not let large volumes of rainfall to infilter. The outputs include natural evaporation and withdrawal by humans. For the last few decades, the balance of water inputs and outputs has become negative, in other words, the volume of water leaving the sea is greater than the amount of water entering the sea, which causes the drying process.

The chemical industry

Due to the special composition of the water, at the southern basin of the Dead Sea potash plants were built. Those plants mine the sea and create evaporation pools which let the water evaporate faster, making use of the local hot and sunny climate, so the firms can use the minerals left behind ("What Can We Do To Stop"). Those minerals are used to produce industrial salts and materials for the cosmetic industry.

Figure 2: map of the streams and rivers in the drainage basin of the Dead Sea



Source: "Map of Israel's Bodies of Water." Bible History

Reduction in Rivers' Water Flow Due to Human Activities

There are two main rivers that deliver water to the Dead Sea, Wadi Mujib (Arnon Stream) and the Jordan River, as can be seen from figure 2. Due to humans, who had built the Degania Dam to keep the water at the Sea of Galilee, a touristic sweet water lake in Israel, and created water storages in Syria, the water flow in the Jordan River has decreased by 98% (Rubinstein). The water is used mainly for drinking and agriculture.

Climate change

The global climate change also has local effects in the area of the Dead Sea, which also contributes to the decreasing water level. It caused the average temperature in Israel to increase by 1.5 °C from 1950 to 2020, but it did not have an effect on the volume of rain during that time (“Climate Change”). The higher temperatures cause increased evaporation, and therefore climate change could also be a reason for the decreasing water level of the Dead Sea.

The importance of conserving the Dead Sea

There are many reasons for why it is important to conserve the Dead Sea. The first reason is that it is a hotspot, a highly biodiverse area that consists of many endemic species, for halophilic microorganisms which are unique to the sea (Green). The drying process has an effect on the freshwater sources in the area which might cause fatal damage to the living organisms in the surrounding ecosystems ("What Can We Do To Stop"). In addition to that, the special features of the Dead Sea and its health value allowed the development of tourism and the economy in the area. The receding shoreline and the formation of sinkholes because of that has caused significant damage to the natural beaches around the sea and to the local tourism that is built on that, which includes hotels, restaurants, and tour guides. The sinkholes also cause economic damage as roads are destroyed and nature reserves cannot be accessed due to the danger for visitors. The economic value of the Dead Sea also comes from the potash plants and the chemical substances mined from the sea. Moreover, the ethical perspective should also be considered. The Dead Sea has an intrinsic value, which is the cultural, aesthetic, or philosophical value of it, regardless of its value for humans (Green). Another ethical issue would be not

keeping the Dead Sea for future generations, so the current generation has a moral obligation to stop the water level from declining any further, and even have to make an effort to restore the water level of the Dead Sea.

Research question and Justification

To what extent do the inputs and outputs of water at the Dead Sea, Israel, allow a sustainable withdrawal of water by the chemical industry from 2000 to 2020?

This time frame was chosen because this is the time when the most significant decline in the water level of the Dead Sea took place. I chose to focus on the sustainable withdrawal of water since those values are not available and so by finding data of all other water inputs and outputs in the system, I will be able to calculate how much withdrawal could be sustainable.

Framework of Investigation

Objectives of the research

My research question aims to find if withdrawal of water from the Dead Sea for the use of the chemical industry (mainly potash plants) can be sustainable, and if it can, then to what extent can the industry withdraw the water. In order to answer this question, I will need to create a model that will balance the water inputs and the water outputs of the Dead Sea. By finding numerical data of the water inputs and outputs, excluding the output due to the current industrial use, I will be able to calculate and show how much extra water is added to the Dead Sea, and based on that, a conclusion of how much water can be withdrawn sustainably by the chemical industry from the Dead Sea could be drawn.

Data collection

In order to create the model, I used secondary data, including newspaper articles, scientific magazines, local tourism websites, governmental databases, and information from experts. To understand the importance, and the sources of inputs and outputs of the Dead Sea, I used newspaper and scientific magazine articles. These were more reliable and useful for this essay especially compared to the tourism websites, which were subjective as they are trying to sell their services, and contained less relevant information. But nonetheless, they highlighted the importance of tourism in the area. The newspaper and magazine look at the sea from different perspectives. The newspaper which I used, Haaretz, gives a more local point of view whereas the scientific magazine, National Geographic, provided me with a global insight. These sources gave me the opportunity to look at the

complex case of the Dead Sea on different scales and from multiple perspectives, which allowed me to understand the actions of different stakeholders in greater depth and to gain general understanding of the system. In addition to that, I also needed to find numerical data of the water inputs and outputs. In order to do this, I used governmental databases, and information from the meteorological and the water services which provided me with the evaporation, precipitation and water level data. However, the flow of water from the river basin to the sea could not be found in those databases, as it is not being measured. Through the help of an expert I contacted, Ittai Gavrieli, who researches geohydrology and the Dead Sea at the Geological Survey of Israel, I was able to get estimations of those values which I used for my model and analysis.

Results and discussion

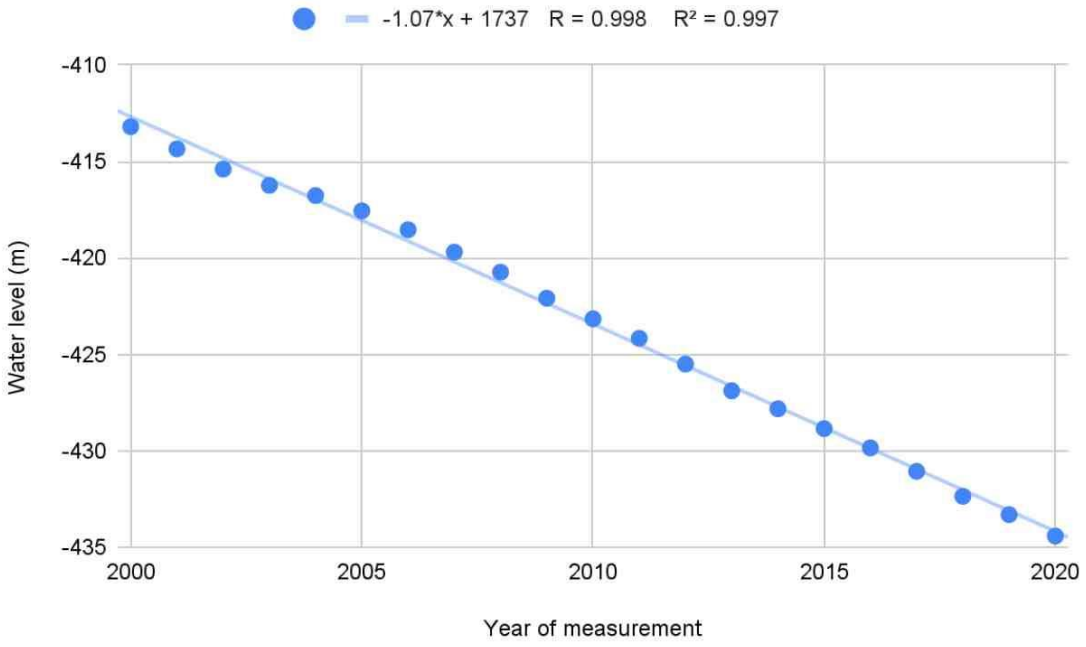
Dead Sea water level

Table 1: The water level of the Dead Sea, from 2000 - 2020

Year	Water Level (m)
2000	-413.200
2001	-414.350
2002	-415.400
2003	-416.240
2004	-416.770
2005	-417.560
2006	-418.540
2007	-419.700
2008	-420.740
2009	-422.090
2010	-423.160
2011	-424.170
2012	-425.503
2013	-426.890
2014	-427.820
2015	-428.850
2016	-429.850
2017	-431.065
2018	-432.360
2019	-433.310
2020	-434.420

Source:"מפלס ים המלח משנת 1976" ["Dead Sea Level from 1976"]. Data Gov

Graph 1: The change of the water level of the Dead Sea per year, from 2000 - 2020



As can be seen from graph 1, there is a strong negative correlation between the year of measurement and the water level of the Dead Sea. The equation of the graph shows an average decrease of 1.07 m per year. There are no outliers in the data, and it stays consistent over the time period between 2000-2020 as the graph is almost fully linear, shown by the correlation coefficient- R which is equal to 0.998 and represents a very strong correlation. This means that there have not been any significant changes in the net water inputs- outputs of the Dead Sea during this time period, the water level is decreasing at a constant rate.

Water outputs from the Dead Sea

Evaporation

Table 2: The average daily and monthly evaporation level in Beit HaArava, Northern shore of the Dead Sea, Israel, from 2000 to 2009 (data was measured only until 2009)

Months	Days in the month	Average daily evaporation (mm)	Average monthly evaporation (mm)
January	31	2.9	89.9
February	28	3.6	100.8
March	31	5.0	155.0
April	30	6.9	207.0
May	31	8.2	254.2
June	30	9.4	282.0
July	31	9.4	291.4
August	31	8.5	263.5
September	30	6.8	204.0
October	31	5.3	164.3
November	30	4.1	123.0
December	31	3.0	93.0

Source: "Evaporation." Israel Meteorological Service.

$Average\ monthly\ evaporation\ =\ days\ in\ the\ month \times average\ daily\ evaporation$

$89.9\ =\ 31 \times 2.9\ (January)$

$Average\ yearly\ evaporation\ level\ =\ sum\ of\ average\ monthly\ evaporation$

$Average\ yearly\ evaporation\ level\ =\ 89.9\ +\ 100.8\ +\ldots +\ 93.0\ =\ 2228.1\ mm\ =\ 2.2281\ m$

$Area\ of\ the\ Dead\ Sea\ =\ 605\ km^2\ =\ 605,000,000\ m^2$

$Average\ amount\ of\ water\ evaporated\ from\ the\ Dead\ Sea\ every\ year\ =\ evaporation\ level\ *\ area$

$Average\ amount\ of\ water\ evaporated\ from\ the\ dead\ sea\ every\ year\ =\ 2.2281\ *\ 605,000,000$

$=1,348,000,500\ m^3\ =\ 1,348\ mcm\ (million\ cubic\ meters/year)$

Water inputs into the Dead Sea

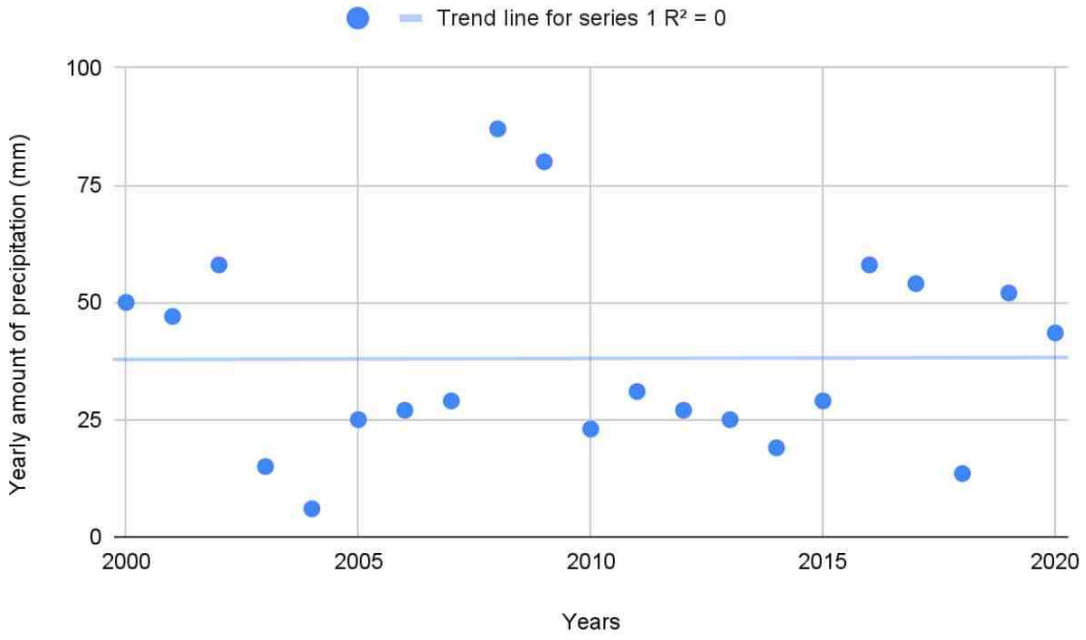
Precipitation

Table 3: The yearly precipitation level at Ein Gedi, Western shore of the Dead Sea, Israel, from 2000 - 2020

Years	Yearly amount of precipitation (mm)
2000	50.0
2001	47.0
2002	58.0
2003	15.0
2004	6.0
2005	25.0
2006	27.0
2007	29.0
2008	87.0
2009	80.0
2010	23.0
2011	31.0
2012	27.0
2013	25.0
2014	19.0
2015	29.0
2016	58.0
2017	54.0
2018	13.5
2019	52.0
2020	43.5

Source: "Climate." Israel Meteorological Service.

Graph 3: The yearly precipitation level at Ein Gedi, Western shore of the Dead Sea, Israel, between the years 2000-2020



As can be seen from Graph 3, there is no correlation between the years and the amount of rain during those years in Ein Gedi, which is located at the western shore of the Dead Sea, as the correlation coefficient is 0. The data points are very far from the trend line, and they do not show any trend or a consistent pattern over time. This means that there hasn't been any change in the average amount of rain during the last 20 years, and during the whole time it is still around the same range that is between 6 mm and 87 mm of rain per year. The average amount of rain per year is 38 mm.

$$Average\ yearly\ precipitation\ level = \frac{sum\ of\ the\ precipitation\ level\ of\ each\ year}{number\ of\ years}$$

$$= \frac{50+47+\ldots+43.5}{21} = 38\ mm = 0.038\ m$$

$$Area\ of\ the\ Dead\ Sea = 605\ km^2 = 605,000,000\ m^2$$

$$Average\ volume\ of\ precipitation\ at\ the\ dead\ sea\ every\ year = precipitation\ level * area$$

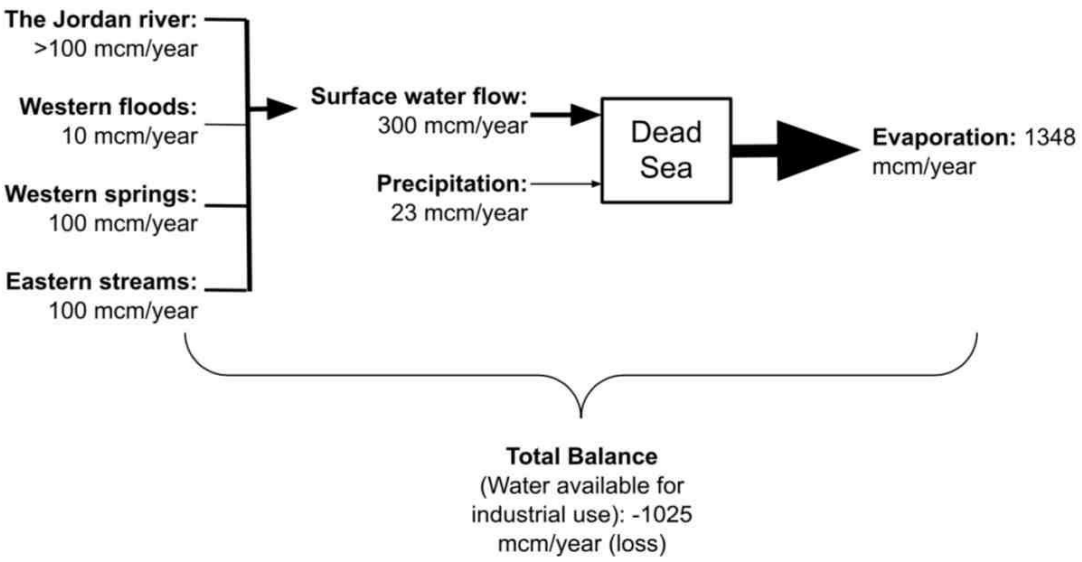
$$= 22,990,000\ m3 = 22.99\ mcm\ (million\ cubic\ meters)$$

Flow of surface water to the Dead Sea

There are only estimations for the volumes of river water flowing into the Dead Sea. There are no measurements taken but the estimations are that slightly less than 100 mcm per year from the Jordan River flow to the Dead Sea (Gavrieli). As a consequence to the floods in the western shore the yearly amount of water is raised by around 10 mcm (Gavrieli). In the past, Eastern streams used to have a flow of around 150 mcm per year, but Jordan has constructed dams on some of the rivers hence the volume today is lower and is about 100 mcm per year (Gavrieli). Another source of water is from western springs (Kane Vasmar and Ein Feshkha) that add just below 100 mcm of water to the Dead Sea per year (Gavrieli). Based on those estimated values, the overall water flow to the Dead Sea is around 300 million cubic metres per year.

Figure 3: the average annual water inputs- outputs balance in the Dead Sea (with the effect of the industrial withdrawal excluded) from 2000-2020

*Model not to scale



Analysis

As can be seen from the figure 3, over 1000 million cubic metres of water from the Dead Sea are lost every year (excluding industrial outputs). This can be explained by the drastic reduction in the volume of surface water flowing into the sea from the rivers in the area. The reduction compared to the past is caused by the diversion of water away from the feeding streams, for domestic and industrial use. The domestic use consists of 85% of the water diverted from the sea, and is mainly used for drinking in Jordan, Israel, Syria, Lebanon, and the Palestinian Authority (ICL). This diversion of water has a political aspect, as it was a part of the peace agreement between Israel, Jordan, Syria and Palestine, showing there are other motives involved ("What Can We Do To Stop"). The water is collected through the construction of dams and the use of water from the reservoirs in the Jordan river and its drainage basin, as well as in Arnon stream over the past few decades. The other 15% are lost because of industrial activity in both Israel and Jordan, which causes a withdrawal of 255 mcm out of the total of 1700 mcm of water diverted from the Dead Sea (ICL). Those 1700 include both domestic and industrial use of water, and therefore are larger than the loss calculated in the model (figure 3). At the beginning of the 20th century, the water inputs and outputs in the Dead Sea were between 1500-2000 mcm/year (volume of evaporated water was larger because the surface area of the sea was bigger), and so they were equal. It is clear that the current diversion of water because of human activities is the cause for the decrease of water level in the sea, and that naturally, the sea could have been in balance.

Conclusion and Application

Conclusion

Overall, as can be seen from the analysis, the water inputs- outputs balance in the Dead Sea, when the withdrawal for industrial use is not taken into account, is an annual loss of 1025 million cubic meters. This means that even without the industrial withdrawal of water from the sea, the balance is still negative. According to graph 1, it would be reasonable to assume that if there is no change made, the water level of the Dead Sea will continue to decrease by roughly 1.1 meters every year and will eventually completely dry. In addition to that, if global warming continues in the future the temperatures in the area will keep increasing, the evaporation level will increase as well. Due to this negative balance, within the current situation, no withdrawal for industrial purposes like potash plants can be sustainable, because even without it, the water level of the Dead Sea is in decline. It is also interesting to see that even though environmentalists tend to blame the chemical industry for the decline in water level, the data shows that the process would have taken place even without it. However, this should not imply that industrial use of water should be allowed, as it will make the situation more severe and will accelerate the rate of decline. If all the diversion of water for domestic use were to stop, and 1445 mcm would have been added to the flow of water to the sea, up to 420 mcm could be used sustainably for the industries of the Dead Sea. However, as the domestic use of water is currently essential for all countries around the drainage basin of the Dead Sea, ending the water diversion would not be possible, unless new significant policies and agreements are put into place.

Possible solutions

To allow the Dead Sea to replenish, a change must be made. Since the main cause for the water level decline of the Dead Sea is the significant reduction in the flow of water, ways to reduce the withdrawal of water from the Jordan river and Arnon Stream should be found. Those could include legalisation on the diversion of water from the drainage basin, however, the diversion of water was a part of the regional peace agreement and therefore any change in the agreement might cause political issues and so political motives are also affecting this decision. Another solution could be to have awareness campaigns in Israel and Jordan, so hopefully less water is wasted in domestic use and as a consequent, less water would need to be extracted from the rivers. Those awareness campaigns have already been done in Israel to some extent, like the “Israel is Drying” campaign that involved well known celebrities in Israel like Ninet Tayeb and Eyal Kitzis, which came together with the introduction of tax on the use of water in households, and reduced the water use by 18% in that month (Bar- Eli). However, a larger scale campaign is needed for better, long lasting results. To reduce the amount of water used, one possibility is to farm with less water. This could be done by farming with crops that require less water like spinach lettuce and beans, or by using more advanced technology that uses water in a more efficient way, like drip irrigation ("Vibrant Gardens"). Another way to reduce the use of water is to expand the use of desalination plants that use water from the sea, which is not scarce, and make it appropriate for domestic and agricultural use. Today, the volume of desalinated water in Israel is 585 mcm per year (Gavrieli). Another solution that will allow the withdrawal of water from the Dead Sea for

industrial use, involves creation of laws by the government. Those laws will charge the industry to give a portion of its profit to develop and fund technologies of water production like desalination, so that the domestic use of water from the rivers will fully stop.

Figure 4: Map of the possible route of the Red Sea- Dead Sea Pipeline



Source: Al-Haditha, Ghor. "Dead Sea's revival with Red Sea canal edges closer to reality."

The last solution that is suggested is the Red Sea- Dead Sea pipeline. This was a plan of the Israeli and Jordanian governments from over ten years ago, in which the idea was to transport water from the Red Sea to the Dead Sea, as can be seen from figure 4, so that it is able to replenish. However, due to criticism from environmentalists, as this solution might damage the special mineral composition of the water and change the condition of the ecosystem and the microorganisms living

in the sea, and due to political and economic issues, this plan has not been done. Again, in many of the solutions politics is a recurring issue and a cooperation between Israel and Jordan is essential in order to save the Dead Sea. From a global perspective, we should tackle climate change by reducing the combustion of fossil fuels and increasing the use of renewable energy, as it will also have an influence on the Dead Sea – increasing temperatures due to climate change also cause more evaporation and speed up the drying process.

Further Research

Further research could be finding the change in water inputs and outputs over a larger period of time. This research has been focused on the last two decades, in which the most significant decline took place. However, in order to find the long term process, and the changes that caused the decline to begin, the research should be expanded and investigate the data from the last 100 years.

Works Cited List

Al-Haditha, Ghor. "Dead Sea's revival with Red Sea canal edges closer to reality." *English*, 18 Mar. 2018, en.prothomalo.com/environment/Dead-Sea's-revival-with-Red-Sea-canal-edges-closer. Accessed 15 July 2022.

Bar- Eli, Avi. "Bar Refaeli Helps Israel Save Water." *Haaretz*, 7 Sept. 2009, www.haaretz.com/2009-09-07/ty-article/bar-refaeli-helps-israel-save-water/0000017f-ec03-d0f7-a9ff-eec75e200000?lts=1657886123723. Accessed 15 July 2022.

"Climate." Israel Meteorological Service, ims.gov.il/en/ClimateAtlas. Accessed 15 July 2022.

Dell'Amore, Christine. "New Life-Forms Found at Bottom of Dead Sea." *National Geographic*, 1 Oct. 2011, www.nationalgeographic.com/science/article/110928-new-life-dead-sea-bacteria-underwater-craters-science. Accessed 10 May 2022.

"Evaporation." Israel Meteorological Service, ims.gov.il/en/evaporation. Accessed 15 July 2022.

Gavrieli, Ittai. "עתיד ים המלח" ["The Future of the Dead Sea"]. *gov.il*, Israel Ministry for Environmental Protection, June 2022, www.gov.il/BlobFolder/publicsharing/dead-sea-future/he/public_comments_2022_dead-sea-draft.pdf. Accessed 15 July 2022.

Green. "Topic 1.1: Environmental Value Systems." *Mr. G Science*, www.mrgscience.com/ess-topic-11-environmental-value-systems.html. Accessed 10 May 2022.

ICL. "Dead Sea Water Level." *ICL Group Sustainability*, ICL, icl-group-sustainability.com/reports/dead-sea-water-level/. Accessed 10 May 2022.

Langfan, Mark. "Topographical Map of Israel." *Israel Science and Technology Directory*, www.science.co.il/israel/Topography.php. Accessed 15 July 2022. Map.

"Map of Israel's Bodies of Water." *Bible History*, bible-history.com/geography/map-palestine-bodies-of-water. Accessed 15 July 2022.

Porat, Sagit. "ים המלח בסכנה" ["The Dead Sea is in Danger"]. *מסע אחר*, www.masa.co.il/article/ים-המלח-בסכנה-חשוב-להציל/. Accessed 10 May 2022.

Ronel, Asaf. "Life Really Can Overcome Anything: How Bacteria Survive 250 Meters Under the Dead Sea." *Haaretz*, 9 Aug. 2019, www.haaretz.com/science-and-health/2019-08-09/ty-article/.premium/life-trumps-all-how-bacteria-survive-250-meters-under-the-dead-sea/0000017f-e16f-d38f-a57f-e77f1e440000. Accessed 15 July 2022.

Rubinstein, Dani. "בלי שתי גדות לירדן" ["Without two banks to the Jordan River"]. *Calcalist*, 10 May 2010, www.calcalist.co.il/local/articles/0,7340,L-3404056,00.html. Accessed 10 May 2022.

"Vibrant Gardens Using Drought Tolerant Vegetables." *Gilmour*, gilmour.com/drought-tolerant-vegetable-garden. Accessed 15 July 2022.

"What Can We Do To Stop The Dead Sea From Dying?" *YouTube*, uploaded by Unpacked, 13 Apr. 2020, www.youtube.com/watch?v=cEDJ3Nm8cR8. Accessed 28 Apr. 2022.

"1976 משנת המלח ים מפלס" ["Dead Sea Level from 1976"]. Data Gov, data.gov.il/dataset/https-www-data-gov-il-dataset-683/resource/823479b4-4771-43d8-9189-6a2a1dcaaf10. Accessed 27 Apr. 2022.

"שינויי אקלים" ["Climate Change"]. *The Israeli Meteorological Service*, May 2022, ims.gov.il/he/node/228. Accessed 15 July 2022.